UNIT 5

Application Layer: Domain name system- DNS Name Space, Resource records, Name Servers. SNMP, Electronic Mail: Architecture and Services, User Agent, Message Formats, Message Transfer, Final Delivery. SMTP, POP3, ICMP. File transfers: FTP, TFTP, Telnet

1 Marks Questions

1. Define **DNS Name Space**.
    - DNS Name Space is a **hierarchical structure** used to organize domain names on the Internet.
2. What is a **Fully Qualified Domain Name (FQDN)**?
    - An FQDN is the complete domain name that specifies its exact location in the DNS hierarchy, e.g., `www.example.com.`
3. What is a **Resource Record (RR)** in DNS?
    - **Resource Records (RRs)** store information about domains. Each RR consists of **Name, Type, Value, TTL**.
4. Name any two **types of DNS Resource Records**.
    - **A** – maps domain to IPv4 address
    - **AAAA** – maps domain to IPv6 address
    - **MX** – mail exchange server
    - **CNAME** – canonical name
5. What is the role of a **Name Server**?
    - A Name Server stores DNS records and **resolves domain names into IP addresses**. **Types of Name Servers:**
        - Root Name Server
        - TLD Name Server
        - Authoritative Name Server
        - Local Name Server

      Name Resolution Process:
      When a client requests a domain name, the query starts from the **local server**, then moves to the **root server**, **TLD server**, and finally to the **authoritative server**, which returns the IP address.

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6. What is **SNMP**? Mention any one use.
    - Simple Network Management Protocol is used to **monitor and manage network devices**.
    - SNMP consists of:
        - **Manager** – monitors network devices
        - **Agent** – runs on managed devices
        - **MIB** – database of managed objects
    - Operations include **GET, SET, GET-NEXT, and TRAP**. SNMP helps in fault detection, performance monitoring, and network management.

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7. Define **Electronic Mail**.
    - Electronic Mail is a service that allows users to **send and receive messages electronically** over a network. Email architecture includes: **User Agent (UA), Mail Transfer Agent (MTA), Mail Delivery Agent (MDA)**
        - **User Agent:** Used to compose and read emails
        - **Mail Transfer Agent:** Transfers mail between servers using SMTP

- **Mail Delivery Agent:** Delivers mail to the user's mailbox

8. What is a **User Agent (UA)**?
   - A User Agent is a software that allows users to **compose, read, reply, and forward emails**.
9. What is **SMTP** used for?
   - S MTP is used for **sending and transferring emails** between mail servers. SMTP is a **push protocol** used to send emails. It operates on **TCP port 25**.
     Steps: Connection → Mail Transfer → Termination.
     Commands include **HELO, MAIL FROM, RCPT TO, DATA, QUIT**.
10. What is the function of **POP3**?
    - POP3 is used by a client to **retrieve emails from a mail server**. Modes:**Delete mode** – mails removed after download or **Keep mode** – mails retained on server.It works on **TCP port 110**.
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11. Expand **ICMP** and state its purpose.
    - ICMP is used for **error reporting and diagnostics**.Common messages:
      - Echo Request / Reply (ping)
      - Destination Unreachable
      - Time Exceeded
12. What is **FTP**?
    - File Transfer Protocol is used to **transfer files between computers** over a network. FTP uses **two connections**: Control connection (port 21) and Data connection (port 20).It supports file upload, download, directory listing, and authentication.
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13. What is **TFTP**?
    - Trivial File Transfer Protocol is a **simplified version of FTP** with no authentication.
14. What is **Telnet**?
    - Telnet allows **remote login** to a system using TCP port 23.
      It enables command execution on a remote machine but is **not secure**, as data is transmitted in plain text.
15. Mention any two **applications of FTP**.
    - Uploading files to a server, downloading software.

Q) Explain the DNS Name Space with a neat diagram.

The **DNS Name Space** is a **hierarchical, tree-structured naming system** used by the Domain Name System to organize and manage domain names on the Internet. It provides a logical way to map **human-readable domain names** to **IP addresses**.

At the top of the hierarchy is the **root domain**, represented by a dot (.). Below the root are **Top-Level Domains (TLDs)** such as `.com`, `.org`, `.edu`, `.gov`, and country-code domains like `.in`, `.uk`. Each TLD may contain **second-level domains** (e.g., `google.com`, `nic.in`), which can further contain **subdomains** (e.g., `mail.google.com`, `www.example.com`).

Each node in the tree represents a **domain**, and every domain has a **unique name** within its parent domain. A complete domain name from a node to the root is called a **Fully Qualified Domain Name (FQDN)**, for example, `www.example.com.`.

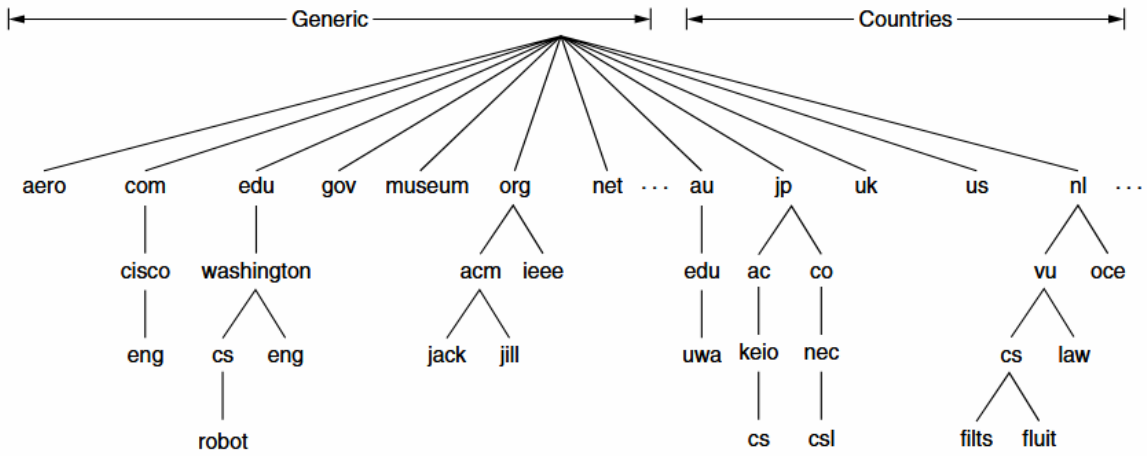


Figure 7-1. A portion of the Internet domain name space.

Each domain controls how it allocates the domains under it. For example,Japan has domains ac.jp and co.jp that mirror edu and com. The Netherlands does not make this distinction and puts all organizations directly under nl. Thus, all three of the following are university computer science departments.

To create a new domain, permission is required of the domain in which it will be included. For example, if a VLSI group is started at the University of Washington and wants to be known as vlsi.cs.washington.edu, it has to get permission from whoever manages cs.washington.edu.

Q) Describe the **types of Name Servers** and explain how DNS name resolution works.

Types of Name Servers

1. **Root Name Server**
   Root name servers are at the top of the DNS hierarchy. They do not store IP addresses of hosts but direct queries to the appropriate Top-Level Domain (TLD) name servers such as `.com`, `.org`, `.in`.
2. **Top-Level Domain (TLD) Name Server**
   TLD name servers manage information for top-level domains. They provide the address of the authoritative name server for a specific domain.
3. **Authoritative Name Server**
   Authoritative name servers store the actual DNS records (A, AAAA, MX, etc.) for a domain. They provide **the** final, correct IP address of the requested host.
4. **Local (Recursive) Name Server**
   A local name server is usually maintained by an ISP or organization. It receives queries from clients, performs recursive resolution, and caches results to improve performance.

DNS Name Resolution Process

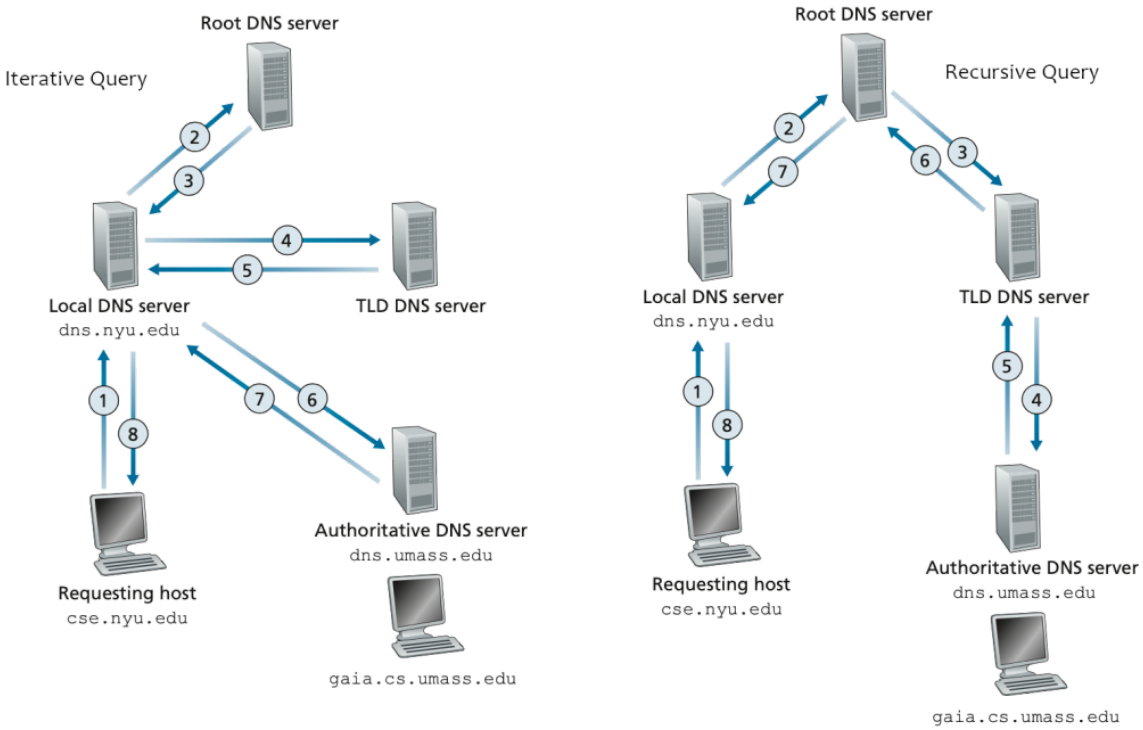
DNS name resolution is the process of converting a **domain name into an IP address**.

Working of Recursive Query

In a **recursive DNS query**, the client expects the DNS server to return the **final IP address** of the requested domain.

1. The client sends a query for a domain name (e.g., `www.example.com`) to the **local DNS server**.
2. If the answer is available in the cache, the local server immediately replies to the client.

3. If not cached, the local DNS server contacts a **root name server**.
4. The root server returns the address of the relevant **TLD name server** (such as `.com`).
5. The local DNS server then queries the **TLD name server**, which responds with the address of the **authoritative name server**.
6. The local DNS server queries the **authoritative name server**, which returns the **IP address** of the requested host.
7. The local DNS server sends the IP address back to the client and **stores it in cache** for future requests.



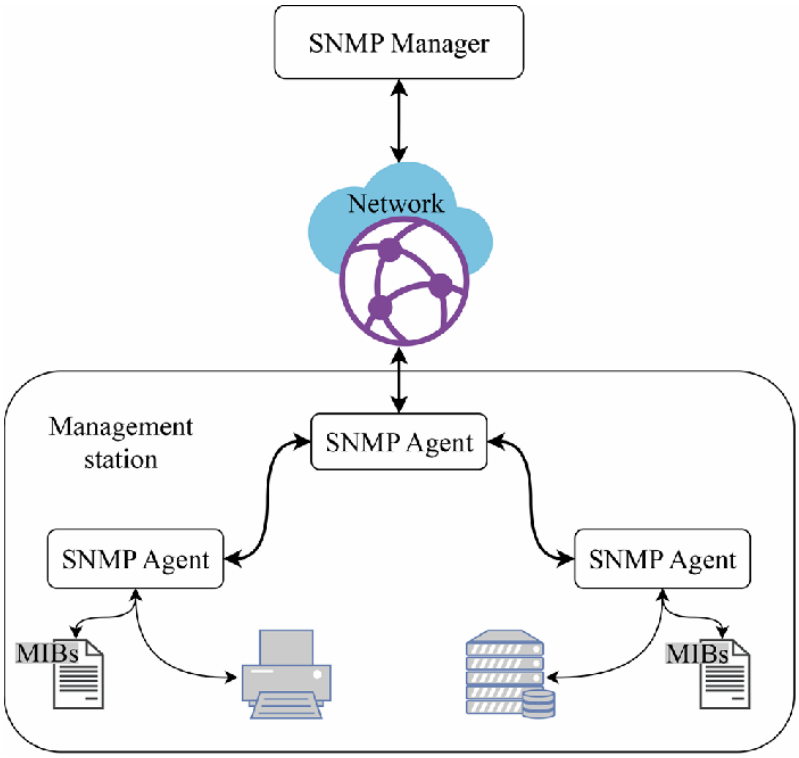
Working of Iterative Query

In an **iterative DNS query**, the client or local server resolves the domain name step by step by querying multiple servers.

1. The client (or local server) sends a query to a **root name server**.
2. The root server does not give the final answer but replies with a **referral** to the appropriate **TLD name server**.
3. The client then queries the **TLD name server** using the referral information.
4. The TLD server responds with the address of the **authoritative name server**.
5. The client queries the **authoritative name server**, which returns the **final IP address** of the domain.
6. The resolved IP address is then used by the client to contact the destination host.

Q) Explain SNMP architecture, components, and operations in detail.

Simple Network Management Protocol (SNMP) is an application-layer protocol used to **monitor, manage, and control network devices** such as routers, switches, servers, and printers.



SNMP Architecture

SNMP follows a **manager–agent model**. A central **SNMP Manager** runs on a Network Management System (NMS) and communicates with **SNMP Agents** installed on managed devices. Each agent maintains information about the device in a **Management Information Base (MIB)**. Communication occurs using SNMP messages over **UDP (ports 161 and 162)**.

SNMP Components

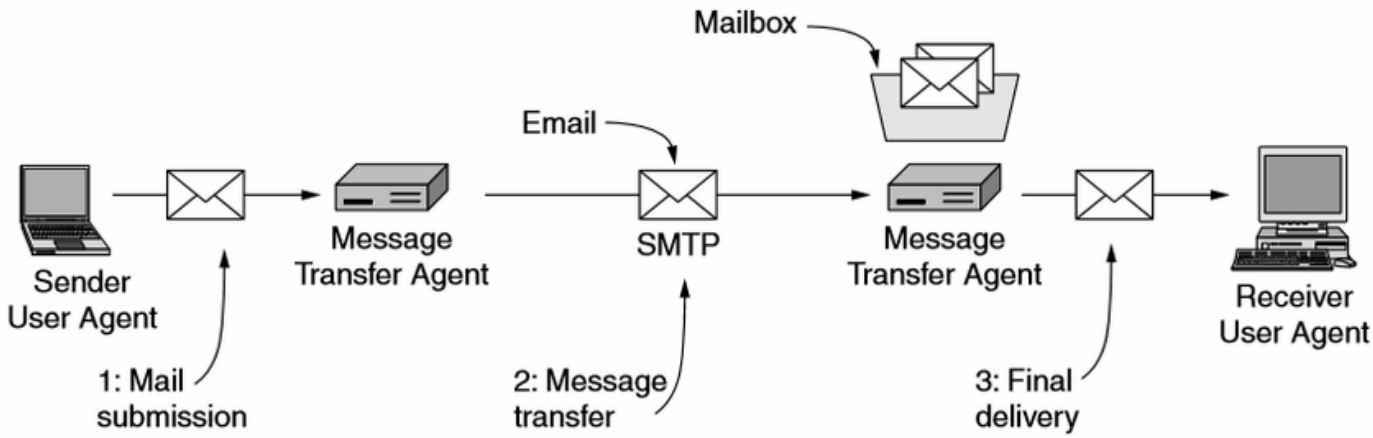
1. **SNMP Manager**
   The manager requests information, sets parameters, and receives alerts from agents.
2. **SNMP Agent**
   The agent collects device data and responds to manager requests.
3. **Management Information Base (MIB)**
   A structured database of managed objects, identified by **Object Identifiers (OIDs)**.
4. **Managed Devices**
   Network elements such as routers, switches, and servers being monitored.

SNMP Operations

- **GET**: Retrieves the value of a managed object.
- **SET**: Modifies the value of a managed object.
- **GET-NEXT / GET-BULK**: Retrieves a sequence of objects.
- **TRAP**: Agent-initiated alert sent to the manager on important events.

Q) Explain the **architecture of Electronic Mail systems** and the **services provided**.

Electronic Mail (E-mail) is an application-layer service that enables users to **send, receive, and manage messages electronically** over computer networks.



Architecture of Electronic Mail Systems

The architecture of an e-mail system consists of the following main components:

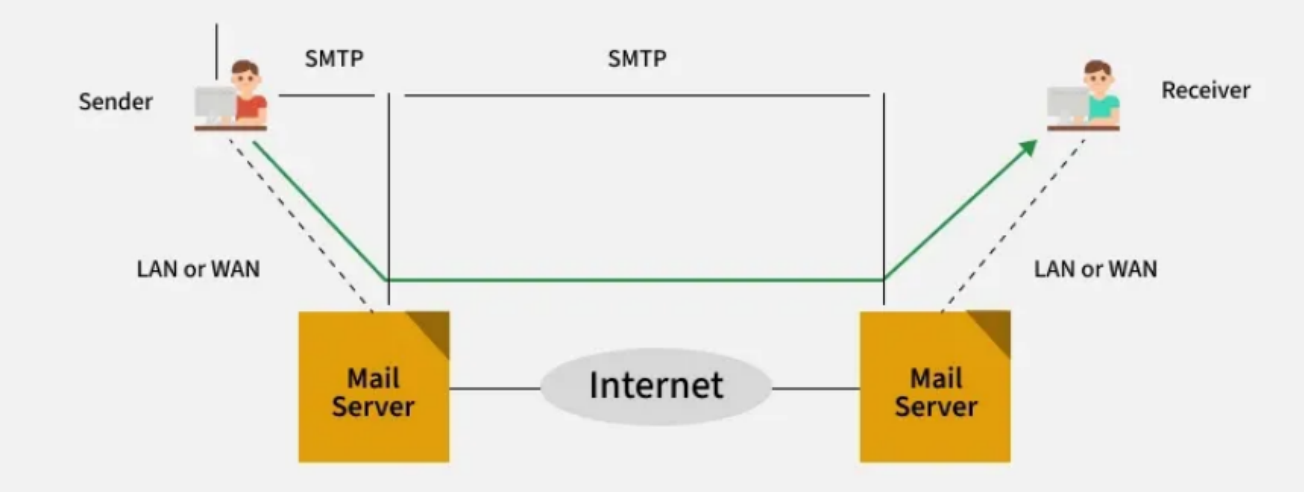
1. **User Agent (UA)**
   The User Agent is the interface between the user and the mail system. It is used to compose, send, read, reply, forward, and delete messages (e.g., Outlook, Gmail).
2. **Mail Transfer Agent (MTA)**
   The MTA is responsible for transferring e-mail messages between mail servers using SMTP. It routes messages from the sender's mail server to the receiver's mail server.
3. **Mail Delivery Agent (MDA)**
   The MDA receives the message from the MTA and delivers it to the recipient's mailbox on the mail server.
4. **Mail Access Agent (MAA)**
   This allows the receiver to access and download e-mails from the server using protocols like POP3 or IMAP**.**

Services Provided by Electronic Mail Systems

1. **Composition** – Creating and editing e-mail messages.
2. **Transfer** – Sending messages from sender to receiver using mail servers.
3. **Reporting** – Notifying the sender about delivery status or failures.
4. **Displaying** – Presenting messages in readable form to the user.
5. **Disposition** – Saving, deleting, replying, or forwarding messages.

Q) Explain SMTP and its working mechanism.

Simple Mail Transfer Protocol (SMTP) is an **application-layer protocol** used for **sending and transferring e-mail messages** between mail servers and from a client to a mail server. It works over **TCP**, usually on **port 25** (port 587 for mail submission).



Working Mechanism of SMTP

SMTP follows a **client–server model** and operates in three main phases:

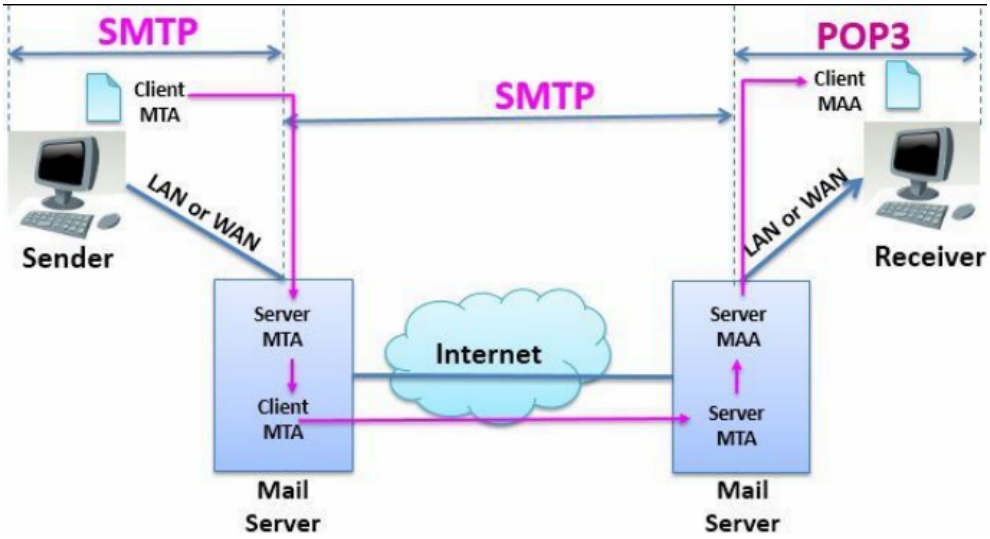
1. **Connection Establishment**
   The sender's mail server establishes a TCP connection with the receiver's mail server. The session begins with commands such as **HELO/EHLO**.
2. **Mail Transfer**
   - The sender identifies the sender and receiver using **MAIL FROM** and **RCPT TO** commands.
   - The actual message is sent using the **DATA** command, which includes the message headers and body.
3. **Connection Termination**
   After successful transfer, the session is closed using the **QUIT** command.

Important SMTP Commands

- **HELO / EHLO** – Initiates communication
- **MAIL FROM** – Specifies sender address
- **RCPT TO** – Specifies recipient address
- **DATA** – Transfers message content
- **QUIT** – Ends the session

Q) Explain POP3 protocol, its modes of operation, and features.

Post Office Protocol version 3 (POP3) is an **application-layer protocol** used by an e-mail client to **retrieve e-mail messages from a remote mail server**. It operates over **TCP**, typically on **port 110** (or port 995 for POP3 over SSL).



Modes of Operation of POP3

1. **Delete Mode**
   In this mode, e-mails are **deleted from the mail server** after being downloaded to the client. This reduces server storage usage and is suitable when mail is accessed from a single device.
2. **Keep Mode**
   In this mode, e-mails are **retained on the mail server** even after download. This allows users to access their mail from **multiple devices**.

Features of POP3

- Simple and easy to implement
- Supports **offline reading** of e-mails
- Uses **client–server architecture**
- Requires **user authentication** (username and password)
- Limited mailbox management (no folder support on server)

Q) Explain IMAP — Internet Message Access Protocol

Internet Message Access Protocol (IMAP) is one of the main protocols used for the **final delivery of e-mail** to the user. **IMAP version 4 (IMAP4)** is defined in **RFC 3501**. In IMAP, the **mail server runs an IMAP server** that listens on **port 143**, while the **user agent acts as an IMAP client**.

Working of IMAP

To use IMAP, the client first **connects to the IMAP server**. If security is required, a **secure transport** (such as SSL/TLS) is started to keep commands and messages confidential. The client then **authenticates** itself using a login mechanism. After successful login, the client can issue various IMAP commands to **list folders, list messages, fetch entire messages or selected parts, mark messages with flags (read, unread, deleted), and organize messages into folders**. Although commonly called *folders*, IMAP refers to them as **mailboxes**, and a user can have **multiple mailboxes** on the server.

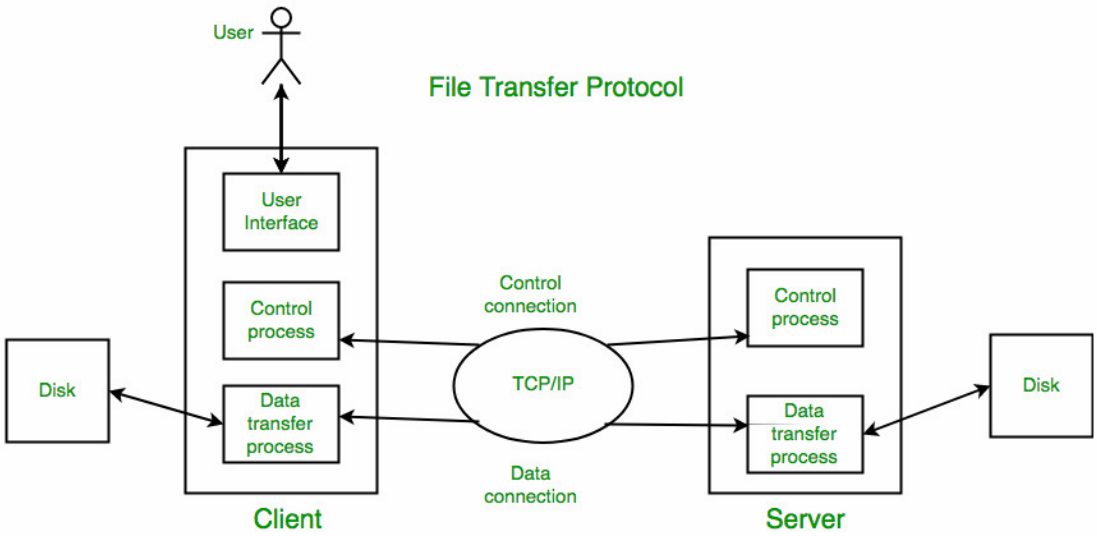
Architecture of IMAP

The Internet Message Access Protocol (IMAP) is a client–server model that allows users to access and view email messages stored on remote servers. Here is a summary of the events:

- **IMAP Clients:** An IMAP client is an email application or software that users use to communicate with their email accounts. Examples include Microsoft Outlook, Mozilla Thunderbird, Apple Mail, and mobile email applications. The client communicates with the server to receive, manage, and send email messages.
- **IMAP Server:** The IMAP server stores email messages and manages user mailboxes. It responds to requests from IMAP clients and provides access to email folders and messages. The server stores emails in a structured format, usually organized in user-defined folders or mailboxes. Common IMAP server software includes Dovecot, Courier IMAP, Cyrus IMAP, and Microsoft Exchange Server.
- **Network Protocols:** It works over TCP/IP (Transmission Control Protocol/Internet Protocol) networks, and allows an client to connect to a server over the Internet or local area networks.

Q) Describe the File Transfer Protocol (FTP) architecture and working.

File Transfer Protocol (FTP) is an **application-layer protocol** used to **transfer files between a client and a server** over a TCP/IP network. It provides reliable file transfer with support for authentication and directory operations.



FTP Architecture

FTP follows a **client–server architecture** and uses **two separate TCP connections**:

1. **Control Connection**

    - Established between the client and server on **TCP port 21**
    - Used for sending **commands and responses** (e.g., login, directory change)
    - Remains open for the entire session
2. **Data Connection**
    - Used for **actual file transfer** and directory listings
    - Opened and closed for each transfer
    - Typically uses **TCP port 20** in active mode

FTP supports:

- **User authentication** (username and password)
- **Anonymous access** (optional)
- **File system operations** such as list, upload, download, rename, and delete

Working of FTP

1. The FTP client establishes a **control connection** with the FTP server.
2. The client **logs in** using valid credentials or as an anonymous user.
3. The client sends commands such as **LIST, RETR (download), or STOR (upload)** over the control connection.
4. For each command involving data transfer, a **separate data connection** is created between the client and server.
5. The requested file or directory data is transferred over the data connection.
6. After the transfer is complete, the **data connection is closed**, while the control connection remains active.
7. The session ends when the client sends the **QUIT** command.

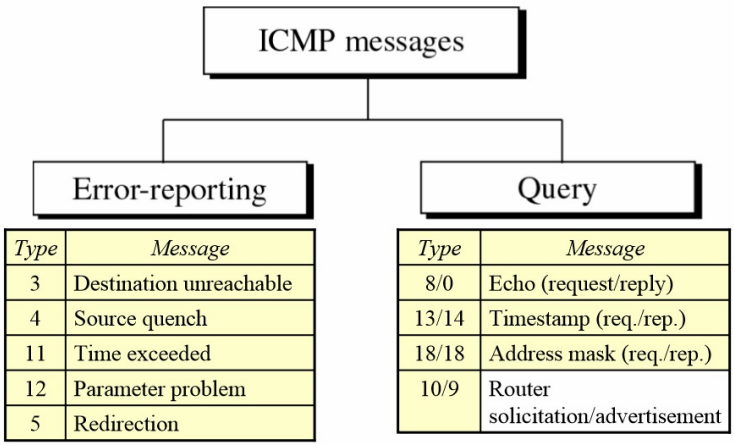
Q) Explain the purpose and functions of **ICMP**.

Internet Control Message Protocol (ICMP) is a **network-layer protocol** used by IP to **report errors and provide diagnostic information** about problems in packet delivery. ICMP does not carry user data; instead, it helps IP operate **reliably and efficiently**.

The main purpose of ICMP is to:

- Report **errors** that occur during IP packet transmission
- Provide **feedback about network conditions**
- Assist in **network diagnostics and troubleshooting**

ICMP messages are generated by **routers or destination hosts** and sent back to the source. **o main categories**: **Error-reporting messages** and **Query messages**. ICMP is used by IP to report problems and provide diagnostic information.



Type	*Message*
3	Destination unreachable
4	Source quench
11	Time exceeded
12	Parameter problem
5	Redirection

Type	*Message*
8/0	Echo (request/reply)
13/14	Timestamp (req./rep.)
18/18	Address mask (req./rep.)
10/9	Router solicitation/advertisement

Error-Reporting Messages: These messages are generated by routers or destination hosts when errors occur during packet delivery. They help the sender understand why a packet could not be delivered.

- **Type 3 – Destination Unreachable**
  Sent when the destination network, host, or port cannot be reached.
- **Type 4 – Source Quench**
  Sent to request the sender to **reduce transmission rate** due to congestion

- **Type 11 – Time Exceeded**
  Generated when the **TTL value becomes zero**, commonly used by *traceroute*.
- **Type 12 – Parameter Problem**
  Indicates an error in the **IP header fields**.
- **Type 5 – Redirection**
  Informs the sender of a **better route** to reach the destination.

2. Query Messages : Query messages are used for **diagnostic and informational purposes** to test reachability and obtain network-related information.

- **Type 8 / 0 – Echo Request / Echo Reply**
  Used by the **ping** command to test host reachability.

- **Type 13 / 14 – Timestamp Request / Reply**
  Used to measure **round-trip time** and time synchronization.
- **Type 18 / 18 – Address Mask Request / Reply**
  Used to learn the **subnet mask** of a network.
- **Type 10 / 9 – Router Solicitation / Advertisement**
  Used by hosts to **discover routers** on the local network.

Q) Write a short note on TFTP and its applications.

Trivial File Transfer Protocol (TFTP) is a **simple, lightweight application-layer protocol** used for transferring files over a network. It operates over **UDP port 69** and provides **basic file transfer functionality** without authentication or directory management.

Features of TFTP (Trivial File Transfer Protocol) – Detailed Explanation

1. **Uses UDP (User Datagram Protocol)**
   TFTP operates over UDP port 69, which makes it simple and fast because there is no connection setup or congestion control as in TCP. However, since UDP is connectionless, reliability is limited. TFTP provides only basic reliability using acknowledgments and retransmissions, but it does not guarantee delivery like FTP.
2. **No User Authentication**
   TFTP does not support usernames or passwords**.** Any client that can reach the server can request a file, which makes TFTP insecure. Because of this, TFTP is normally used only in trusted environments, such as local area networks.
3. **Supports Only Read and Write Operations**
   TFTP supports only two basic operations:
   - RRQ (Read Request)**:** to download a file from the server
   - WRQ (Write Request): to upload a file to the server
     It does not support commands like file rename, delete, or directory navigation.
4. **Limited Error Handling and No Directory Listing**
   TFTP has very simple error handling. It can report basic errors such as file not found or access violation, but it cannot handle complex error recovery. Also, TFTP does not allow directory listing, making file management very limited.
5. **Best Suited for Small File Transfers**
   TFTP transfers data in small fixed-size blocks (usually 512 bytes). Due to its simplicity, lack of security, and limited functionality, it is mainly used for small files such as configuration files, boot files, and firmware images.

Working of TFTP (Brief)

A TFTP client sends a **read or write request** to the TFTP server. Data is transferred in **fixed-size blocks**, and each block is acknowledged before the next one is sent. If an acknowledgment is not received, the block is retransmitted.

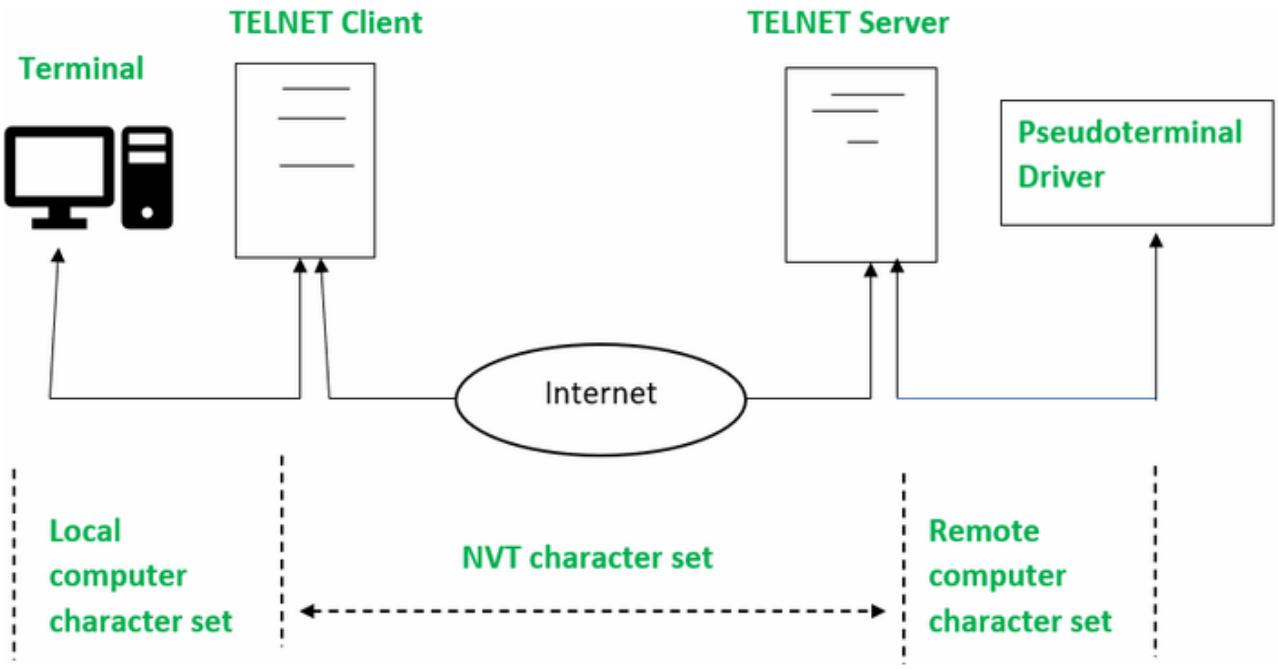
Applications of TFTP

1. **Network booting** of diskless workstations
2. **Loading firmware** or configuration files into routers and switches
3. **Backing up configuration files** of network devices

4. Simple file transfers in **trusted local networks**

Q) Explain Telnet and its working principle.

Telnet is an **application-layer protocol** used for **remote login** into another computer over a TCP/IP network. It allows a user to access a remote system and execute commands as if working on the local machine. Telnet operates over **TCP port 23**.



Working Principle of Telnet

Telnet follows a **client–server model**:

1. The **Telnet client** initiates a TCP connection to the **Telnet server** on port 23.
2. Once the connection is established, the server prompts the client for **username and password**.
3. After successful authentication, a **virtual terminal** is created on the client system.
4. The user's keystrokes are sent to the remote server as **plain text**.
5. The server executes the commands and sends the output back to the client.
6. The session continues interactively until the user logs out or terminates the connection.

Features of Telnet – Detailed Explanation

1. **Supports Remote Command Execution**
   Telnet allows a user to log in to a remote computer and execute commands as if the user were physically present at that system. Once authenticated, the remote machine accepts commands from the user and returns the results. This feature is useful for remote administration, configuration, and troubleshooting of network devices and servers.
2. **Uses Network Virtual Terminal (NVT) Concept**
   Telnet uses the Network Virtual Terminal (NVT) mechanism to ensure interoperability between different hardware and operating systems. The client converts local keyboard input into a standard NVT format before sending it, and the server converts NVT data into its local format. This allows communication between heterogeneous systems without compatibility issues.

3. **Simple and Easy to Use**
   Telnet is a lightweight protocol with minimal overhead. It requires no complex setup and provides a straightforward text-based interface, making it easy to implement and use. Due to its simplicity, Telnet was widely adopted in early computer networks.
4. **Unencrypted Plain Text Transmission**
   All data in Telnet—including usernames, passwords, and commands—is transmitted as plain text over the network. This makes Telnet vulnerable to eavesdropping and security attacks such as packet sniffing. Because of this major security weakness, Telnet is now largely replaced by Secure Shell (SSH) in modern networks.